RIP Setup

BY

Prasad Nikam

Prasad Nikam

Introduction

Routing Information Protocol (RIP) is a distance-vector routing protocol that enables routers to exchange routing information and determine the best path for forwarding data packets. It uses hop count as its routing metric, selecting the route with the fewest router hops. In this practical setup, RIP is configured in Cisco Packet Tracer using five routers, a PC, and a server to demonstrate how dynamic routing enables communication between different networks.

RIP

Routing Information Protocol (RIP) is a network protocol that routers use to find the best path for data based entirely on hop count (the number of routers a packet must pass through).

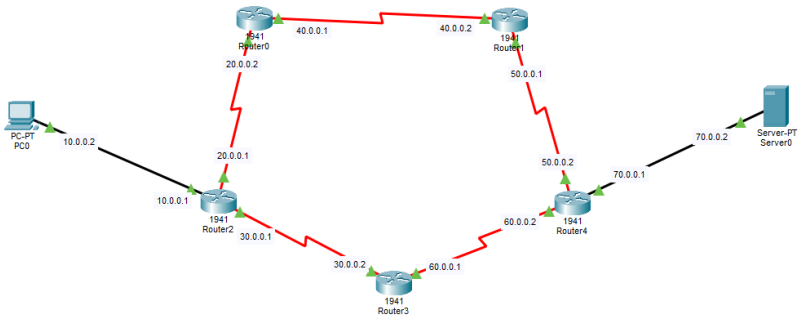
Here is how it works in short:

- Table Sharing: Every 30 seconds, routers broadcast or multicast their entire routing table to their neighbours.
- Path Selection: If a router discovers a path with fewer hops to a destination, it updates its table to use that shorter route.
- Network Limit: The maximum allowed distance is 15 hops. A distance of 16 hops means the destination is unreachable, which prevents endless routing loops but limits RIP to smaller networks.

Routing Information Protocol (RIP) is heavily limited by its 15-hop maximum distance, which completely restricts its use to small networks and treats any destination 16 hops away as unreachable. Furthermore, RIP selects paths based solely on hop count rather than bandwidth, meaning it will stubbornly route traffic through a painfully slow link if it has fewer routers than a high-speed alternative. Because it constantly broadcasts its entire routing table every 30 seconds, it consumes excessive network bandwidth and suffers from slow convergence, causing lengthy delays and dangerous routing loops whenever a network link fails.

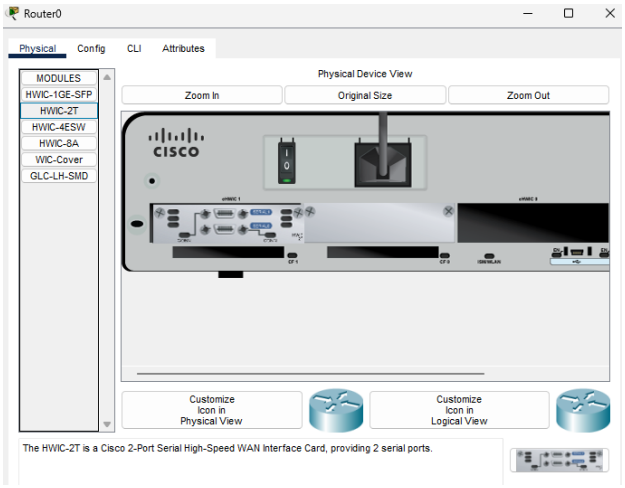
RIP Setup

For a basic understanding of RIP works, we need a PC, 5 routers and a server. Arrange them like shown following:

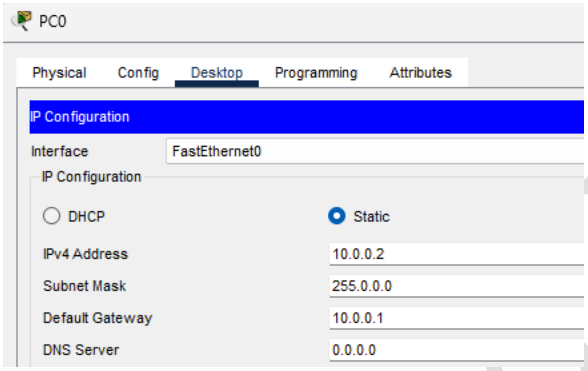


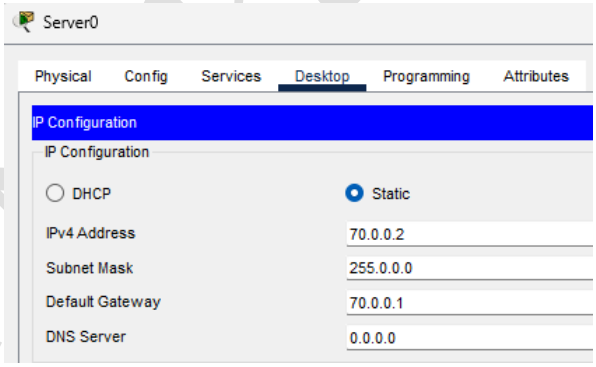
Now we need to assign each component in the setup IP for them to communicate with each other. Remember to assign IP according to the particular network as shown in the SS. Why? Because each router connecting with other is a different network, so for them to communicate with each other they must share an IP of the same series. Like if router1 is connected to router0 then for that communication we need a specific IP series. While doing connection you might get an error that there is no port available as a router only has one fast ethernet port, so we need to add port to that open your router and turn it off and the following port (HWIC-2T):

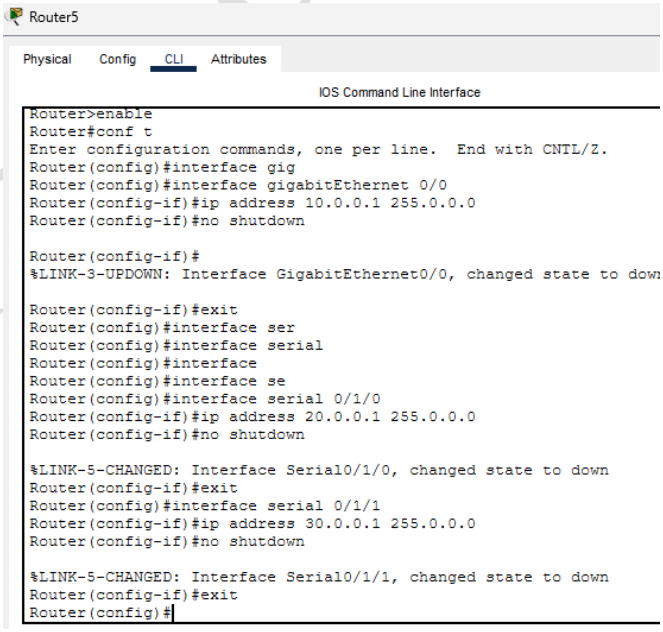
Prasad Nikam



After doing so we have 2 more additional ports which are serial. So, we need to connect the router with each other via serial DCE cable. Now we need to configure the IP for each device i.e. PC, server and routers.



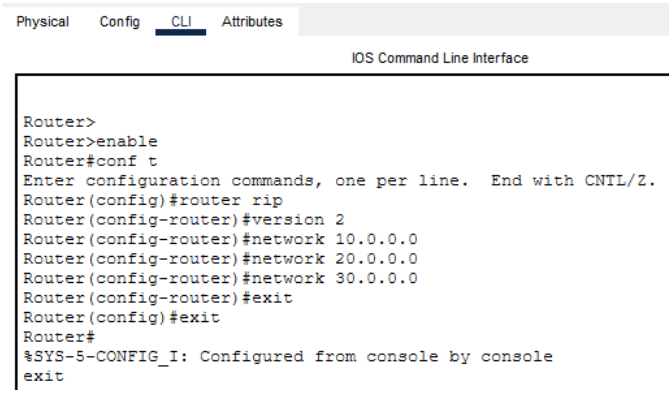




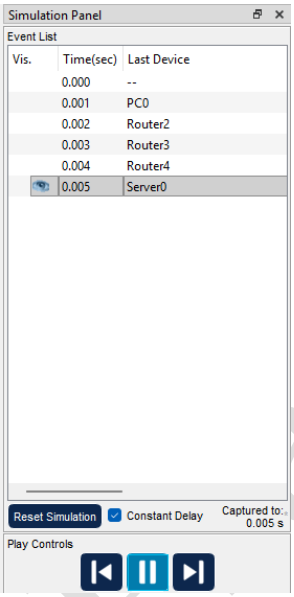
Now we are done with the connection setup, but if we send a packet from PC to the server, you will see failed message. Because we have not made route for it travel from one router to another, so to send the packets we need to enable routing.

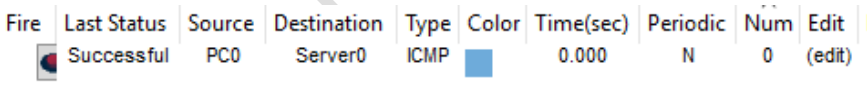
RIP Configuration

Now to set up communication from PC to Server, we need to configure the RIP protocol. To do so we need to open CLI in router and assign each network connected to that router in the RIP protocol.



Here the router2 has networks of series 10, 20, and 30. We need to do so for each router. This will enable the protocol path for each router and find the shortest path for each communication between this network. So now if we send a packet from PC0 to Server0, it will travel through the shortest path possible, i.e. PC0 → router2 → router3 → router4 → server0. Let's check this in packet tracer:





Conclusion

The RIP setup was successfully configured in Cisco Packet Tracer, enabling communication between the PC and the server through multiple routers. By advertising their connected networks and exchanging routing tables, the routers dynamically learned the available routes and selected the path with the lowest hop count. The successful packet transmission demonstrated how RIP simplifies routing between interconnected networks without requiring manual static routes. However, its 15-hop limit and reliance on hop count make it more suitable for small networks.

Fasaq Nikam